

MARKING SCHEME

Class X | Biology | Unit Test II | Maximum Marks: 50

Award marks for each value point shown. Accept equivalent wording. Spelling is not penalised. For diagram questions, labels must be on a drawn diagram.

Q.	Value points	Marks
1. [2]	Arteries carry blood away from the heart.	1
	Names the aorta as the main (largest) artery.	1
2. [2]	Chloroplast.	2
3. [2]	Names chlorophyll a and chlorophyll b.	1
	Names the two stages: light reaction and dark reaction (Calvin cycle).	1
4. [2]	Right atrium to right ventricle through the tricuspid valve.	0.5
	Right ventricle to the lungs through the pulmonary valve and pulmonary artery.	0.5
	Pulmonary veins to left atrium, then left ventricle through the bicuspid (mitral) valve.	0.5
	Left ventricle to the aorta through the aortic valve.	0.5
5. [2]	Diagram of an alveolus is drawn.	0.5
	Labels the alveolar sac.	0.5
	Labels the capillary.	0.5
	Arrows show oxygen into the blood and carbon dioxide out.	0.5
6. [5]	Neat diagram of the human digestive system is drawn.	1
	Labels stomach, small intestine and large intestine.	1
	Labels liver and pancreas.	1
	Marks the small intestine (ileum) as the site of most absorption.	1
	Mentions villi increasing the surface area for absorption.	1
7. [5]	Diagram of a nephron is drawn.	1
	Labels Bowman's capsule and glomerulus.	1
	Labels the proximal tubule.	1
	Labels the loop of Henle.	1
	Labels the collecting duct.	1
8. [5]	Palisade cells are elongated or column shaped.	1
	Palisade cells are tightly packed.	1
	Palisade cells contain many chloroplasts to absorb the most light.	1
	Spongy cells are irregular and loosely packed.	1
	Large air spaces in the spongy layer help gas exchange.	1
9. [5]	Defines transpiration as the loss of water vapour from the aerial parts of a plant.	2
	States that it happens mainly through the stomata.	1
	Names one factor that increases the rate (high temperature, wind, low humidity or bright light).	1
	Names a second such factor.	1

Q.	Value points	Marks
10. [5]	Xylem vessels are dead, hollow tubes with no end walls.	2
	This forms a continuous column for water to move up.	1
	Lignified walls give strength and stop the vessels collapsing.	2
11 (a). [2]	States that Plant A is healthier.	1
	Reason: broad green leaves show more chlorophyll and good light.	1
11 (b). [3]	Move Plant B into bright light or sunlight.	2
	Reason: light is needed for photosynthesis and chlorophyll formation.	1
12. [5]	Formula: minute ventilation = tidal volume x breathing rate.	2
	Correct substitution: 0.5×12 .	1
	Correct answer with unit: 6 L per minute.	2
13. [5]	Formula: alveolar ventilation = (tidal volume - dead space) x rate.	2
	Correct substitution: $(0.5 - 0.15) \times 12$.	1
	Correct answer with unit: 4.2 L per minute.	2
	Total	50